I. USING THE ALREADY EXISTING DATABASE

```
USE employee;
```

II. ENTERING DATA INTO TABLES OF THE DATABASE

```
INSERT INTO departments (department_id, department_name) VALUES
 (1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
(7, 'Finance'),
(8, 'Design'),
(9, 'Research and Development'),
(10, 'Customer Support'),
(11, 'Business Development'),
(12, 'IT'),
(13, 'Operations');


INSERT INTO location (location) VALUES
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');


INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation,
department_id, location_id, salary) VALUES
(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),
(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
```

(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),

(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),

(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),

(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),

(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),

(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),

(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),

(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),

(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),

(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),

(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),

(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),

(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),

(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),

(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),

(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),

(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),

(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),

(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),

(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),

(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),

(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);

III. DISTINCT VALUES:

SELECT DISTINCT salary FROM Employees;

IV. ALIAS (AS):

SELECT age AS Employee_Age,salary AS Employee_Salary FROM Employees;

V. WHERE CLAUSE & OPERATORS:

SELECT * FROM Employees WHERE salary > 50000 AND hire_date < '2016-01-01';

SELECT * FROM Employees WHERE designation = 'Data Scientist';

VI. SORTING AND GROUPING DATA:

1. **ORDER BY:**

   SELECT * FROM Employees ORDER BY department_id ASC, salary DESC;

2. **LIMIT:**

SELECT * FROM Employees WHERE hire_date >= '2018-01-01' AND hire_date < '2019-01-01'
ORDER BY hire_date ASC LIMIT 5;

3. **AGGREGATE FUNCTIONS:**

- SELECT SUM(e.salary) AS Total_Finance_Salary

  FROM Employees e

  JOIN Departments d

  ON e.department_id = d.department_id

  WHERE d.department_name = 'Finance';

- SELECT MIN(age) AS Minimum_Age
  FROM Employees;

4. **GROUP BY:**

- SELECT l.location,

  MAX(e.salary) AS Maximum_Salary

  FROM Location l

  LEFT JOIN Employees e

  ON l.location_id = e.location_id

  GROUP BY l.location_id, l.location;

- SELECT designation,

  AVG(salary) AS Average_Salary

  FROM Employees

  WHERE designation LIKE '%Analyst%'

  GROUP BY designation;

5. **HAVING:**

- SELECT d.department_name,

  COUNT(e.employee_id) AS Total_Employees

  FROM Departments d

  LEFT JOIN Employees e

  ON d.department_id = e.department_id

  GROUP BY d.department_id, d.department_name

  HAVING COUNT(e.employee_id) < 3;

- SELECT l.location,

  AVG(e.age) AS Average_Age

  FROM Location l

  JOIN Employees e

  ON l.location_id = e.location_id

  WHERE e.gender = 'F'

  GROUP BY l.location_id, l.location

  HAVING AVG(e.age) < 30;

VII. JOINS:

1. **INNER JOIN:**

   SELECT e.employee_name,

   e.designation,

   d.department_name

   FROM Employees e

   INNER JOIN Departments d

   ON e.department_id = d.department_id;

2. **LEFT JOIN:**

   SELECT d.department_name,

   COUNT(e.employee_id) AS Total_Employees

   FROM Departments d

   LEFT JOIN Employees e

   ON d.department_id = e.department_id

   GROUP BY d.department_id, d.department_name;

3. **RIGHT JOIN:**

   SELECT l.location,

   e.employee_name

   FROM Employees e

   RIGHT JOIN Location l

   ON e.location_id = l.location_id;